Sindhudurg Shikshan Prasarak Mandal's College of Engineering, Kankavli

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Experiment No -6		
Year: 2026-27	Class/Div.-TE	SEM: V
Course Code: 2115311	Course Name: Open Source Technology	
Branch: Computer Engineering, Computer Science & Engineering & AIML (Computer Science Engineering)		

Aim	**Linux Shell Mastery** Execute systemnavigation using terminal tools (cd, ls, pwd); Process large files using standard automation filters (grep, awk, cat, |).
Software Requirements	**OS:** Any Linux distribution (Ubuntu, Debian, Fedora, Arch), macOS, or WSL. **Terminal:** GNOME Terminal, Konsole, Alacritty, or default OS terminal. **Core Utilities:** Standard GNU coreutils (cd, ls, pwd, cat), grep, and awk (pre-installed).
Theory	**A. System Navigation & Directory Management** The Linux file systemisstructuredasasinglehierarchicalinvertedtreerootedat/. Everything (files,directories,devices,processes)isrepresentedwithinthistree. pwd**(Print Working Directory):** Outputstheabsolutepathofthe current directory, serving as your spatial reference point in the file system tree. cd **(Change Directory):**Navigatesbetweennodesinthedirectory hierarchy. Accepts both absolute paths (starting from /) and relative paths (starting from current working location). Special relative shortcuts include . (current directory), .. (parent directory), ~ (user's home directory), and - (previous directory). ls**(List):**Displaysdirectorycontents.Standardflagsinclude-l(longlisting format showing permissions, ownership, size, and timestamps), -a (includes hidden files starting with .), and -h (human-readable file sizes). **B. Text Processing & Data Pipelines** Linuxhandles datastreamsusing standardstreams:**stdin**(filedescriptor0), **stdout** (file descriptor1),and **stderr**(filedescriptor2). cat (**Concatenate**):Reads files sequentially andwrites themtostandard output. Useful for viewing small files or combining multiple files. grep (**Global Regular Expression Print**):Scansinput line-by-linefor pattern matches based on regular expressions and prints matching lines. Flags include -i (case-insensitive), -v (invert match), and -c (count matching lines). awk(**Aho, Weinberger, and Kernighan Pattern Scanning Language**): A domain-specific programming language designed for

	field-oriented text processing. It parses data row-by-row into fields delimited by whitespace or custom field separators (FS / -F), referring to column fields as $1, $2, ... $N, and the entire record as $0. | (**Pipe Operator**): Connects the standard output of one process directly to the standard input of another process in memory, allowing complex data processing pipelines without writing temporary files to disk.
Procedure	**Step 1: System Navigation Practice** **1.Verify Working Location:**Initial spatial reference. Determiney ourcurrentpathinthe filesystem hierarchy: pwd **2.Explore Directory Contents:**List detailed metadata. List all contents in the root directory, including hidden files and human-readable file sizes: ls -lah / **3.Navigate Path Hierarchies:**Absolute and relative traversal. Move to /var/log using an absolute path, then step up two levels using relative references: cd /var/log pwd cd ../.. pwd **Step 2: Sample Dataset Generation** To practice stream filtration, generate a sample log file named server.log: cat << 'EOF' > server.log 2026-08-31 08:00:01 INFO User admin logged in successfully 2026-08-31 08:05:12 ERROR Database connection timeout on port 5432 2026-08-31 08:06:45 WARN Disk usage above 85 percent 2026-08-31 08:12:03 ERROR Failed login attempt from IP 192.168.1.45 2026-08-31 08:15:22 INFO Scheduled backup completed 2026-08-31 08:20:11 ERROR Database query failed: syntax error 2026-08-31 08:22:00 WARN High memory usage detected EOF **Step 3: Text Filtration with Pipelines** **1.Display Log File Contents:** Verifyr aw data stream. Streamtheentirerawlogfiletostandard output: cat server.log

	2.Filter Error Messages: Pattern match with grep. Isolate lines containing the term ERROR (case-insensitive): grep -i "error" server.log **3.Extract Specific Fields via AWK:**Field processing. Extract the timestamp (columns 1 & 2) and log level (column 3) for all log entries: awk '{print $1, $2, "-->", $3}' server.log **4.Combine Tools via Pipeline:** Multi-stage filter. Chain grep and awk using a pipe (|) to extract only the timestamps of all critical ERROR events: grep "ERROR" server.log | awk '{print $1, $2, "Error Detail:", $0}'
Conclusion	This experiment verified the core fundamentals of Linux terminal interaction. Essential system navigation commands (cd, ls, and pwd) establish total spatial control over both relative and absolute file paths within the Unix directory tree. Furthermore, leveraging stream composition with the pipe operator (|) allows modular CLI utilities to be chained together into dynamic, memory-efficient analytical workflows without generating temporary files. Finally, combining targeted pattern matching via grep with field extraction through awk enables rapid, flexible data filtering and structured reporting directly from raw text logs.

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